

No Free Credit: How Much Simulated Evidence Can Replace Real Events in Zero-Failure Reliability Demonstration of Perception Systems

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Abstract

A perception system on a new fixed camera is often accepted by a zero-failure reliability demonstration: a fixed number of real events, the zero-failure sample size (59 events for 95% confidence that at most 5% of events are missed), must pass without a miss. How many can simulated evidence replace? We prove that, without an assumption linking the simulator to reality, independent simulated data cannot shorten the demonstration; that accepting a larger error probability buys an exactly computable number of events; and that a synthetic twin paired with each real event bounds the real miss rate by the twin miss rate plus a discordance rate, which pays off only through a transfer bound calibrated across scenes and priced by the number of scenes. On 78 fixed-camera scenes, real misses are rare at the operating point fixed in advance (0.7%), so the certificate is untested at that operating point. Even if the calibration carried over to new cameras, a twin run about 27 times per camera would save at most 14 events per camera at a 20% target; the twin as built saves none. A named failure mode, out-of-domain pass-through, remains: judged alone, 4 of 6 night cameras are falsely certified.

Keywords: reliability demonstration, zero-failure testing, synthetic data, digital twin, success-run, perception system, transfer bound

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1. Introduction

A reliability demonstration test (RDT) accepts a product when it passes a planned test; for attribute (pass/fail) data the classical plan is the zero-failure, or success-run, test [1, 2]. If each demand fails independently with probability p , a unit with $p > \varepsilon$ passes n demands with probability at most $(1-\varepsilon)^n$, so passing $n_0(\varepsilon, \delta) = \lceil \ln \delta / \ln(1-\varepsilon) \rceil$ demands without failure demonstrates $p \leq \varepsilon$ at confidence $1 - \delta$: 59 demands at $\varepsilon = \delta = 5\%$. Zero-failure planning remains common industrial practice [3], and its sample-size burden is a recurring motive for test-plan research in this journal, from success-failure verification tests [4] to qualification plans that borrow auxiliary information [5, 6].

We consider a learned perception component, a fixed object detector on a fixed camera that samples frames periodically, as the unit under test. A demand is the passage of an object (an *event*); a failure is an event on which no sampled frame yields a detection. Such components are increasingly assessed with reliability-engineering tools [7, 8, 9]. For them, the demands are rare and every new installation is a new unit: 59 failure-free events must be collected per camera, and for road vehicles the exposure needed to demonstrate rare-failure reliability by field operation alone is prohibitive [10].

Simulation is the obvious remedy. Digital twins and simulators already supply reliability evidence in engineering practice [11, 12, 13], and virtual scenario testing is a leading proposed alternative to mileage-based demonstration of automated vehicles [14, 15]. For perception, however, the simulation-to-reality gap is measurable and has been measured under different environmental conditions [16, 17], and performance in simulation need not predict performance in the field [18]. The question for a demonstration test is therefore quantitative: *how many of the n_0 real demands can simulated evidence replace, and what form of guarantee survives when the evidence is transferred to an untested unit?*

This paper answers the question for zero-failure demonstration of a perception component.

1. *No free credit.* Without an assumption that links the simulator to reality, no valid procedure can use independent simulated data to shorten a zero-failure demonstration, for any simulator and any amount of simulation. This makes precise, for success-run tests, a principle known from clinical borrowing [19], and relates to the limits on combining evidence in software dependability [20].

2. *The exact price of relaxing confidence.* Accepting error probability $\delta' > \delta$ when the simulator is wrong buys at most, and exactly, $n_0(\varepsilon, \delta) - n_0(\varepsilon, \delta')$ demands: 8, 14, 22 and 27 of the 59 for $\delta' = 7.5, 10, 15$ and 20%.
3. *A paired-twin bound and a fleet-level transfer bound.* A synthetic twin of each real event bounds the real failure probability by the twin failure probability plus a discordance rate, with no assumption; on one camera this costs more real events than the plain test (72–122). Calibrated once across scenes, the discordance transfers to an untested camera as a population-average bound whose price is the number of calibration scenes, not the number of events.
4. *An exchange rate on public benchmarks, and a named failure mode.* On 78 fixed-camera scenes the twin as built saves no real event; a twin run to convergence would save at most 14 real events per camera, if the discordance calibrated with paired twins also held for the twins of new cameras (not verified), at the operating point fixed in advance, where real failures are rare (0.73%) and the demonstration is therefore untested. When out-of-domain (night) cameras are judged alone, 4 of 6 are falsely accepted; we call this *out-of-domain pass-through*.

Reliability demonstration tests in the zero-failure regime have no established rule for how much simulated evidence may substitute for real evidence, nor a guarantee form that survives transfer to an untested unit; this paper gives a finite-sample answer (none without an assumption; an exact price for relaxing confidence; a fleet-level bound whose price is scene count) and measures it on 78 fixed-camera scenes.

2. Related work and research gap

Table 1 positions the paper against four bodies of work.

(a) *Reliability demonstration and zero-failure test planning.* The zero-numerator bound and the rule of three are classical [1, 2], and the Bayesian zero-failure (BAZE) test sets the required test time through a prior on the failure rate [21]. Recent work designs sample sizes by assurance [22], studies the limits of Bayesian zero-failure plans [3], and publishes in this journal demonstration and acceptance plans for degrading products [23, 34], success-failure verification tests [4] and qualification plans that use subsystem, Monte-Carlo or expert information [5, 6]. For software and autonomous

Table 1: Research gap. Unit: what the guarantee covers. ZF: is the zero-failure (success-run) regime treated? Price: is the real-evidence value of simulation quantified? Transfer: does the guarantee reach an untested unit? Link: assumption needed between simulation and reality. Evidence: empirical basis.

Body of work	Unit	ZF	Price of simulation	Transfer	Sim-real link	Evidence
(a) Reliability demonstration, zero-failure planning [1, 21, 22, 23, 4, 5, 24]	product / unit	yes	through a prior or auxiliary data; not as real events saved	to the tested product; prior-system evidence handled conservatively	prior or similarity assumption (Bayesian); none (frequentist, no simulation)	case studies
(b) Simulation-assisted testing, digital-twin reliability [11, 12, 13, 25, 26]	system	no	fidelity weighting in fusion	not addressed (extrapolation named as open)	fidelity or calibration model	engineering case studies
(c) Simulation-augmented evaluation of perception [16, 17, 27, 15, 28, 29]	mean metric	no	gap measured; variance reduction for means	no	paired labels (rectifier) or none	perception benchmarks
(d) Finite-sample guarantees and assurance for learned perception [30, 31, 32, 33, 8]	per deployment	no	not addressed (real calibration data)	exchangeability of calibration and deployment	exchangeability	detection data sets
(e) This paper	fleet (population average); per-unit as a data requirement	yes	exact: none without a link; $n_0(\delta) - n_0(\delta')$ when relaxing; 14 events per camera at most	yes, priced by scene count ($U(J, m) = 5.94\%$, $J = 1$, $m = 78$)	none for the twin bound; scenes exchangeable (i.i.d.) for transfer, as in (d) at scene level	78 fixed-camera scenes, 180 events; failure mode measured

73 systems, conservative Bayesian inference uses prior knowledge, including ev-
74 idence from previous similar systems, without optimistic bias [24, 35, 36],
75 and reusing the reliability of an older system as a prior is optimistic rather
76 than conservative [37]. Bayesian borrowing quantifies how much a prior is
77 worth through its effective sample size [38] and discounts historical data
78 through power or commensurate priors [39, 40], and conservative frequentist
79 bounds have been derived when the test profile differs from the operational
80 profile [41]. *What is missing:* a frequentist statement of how much the zero-
81 failure sample size can be reduced by simulated rather than prior-system
82 evidence, and at what price.

(b) *Simulation-assisted testing and digital-twin reliability.* Digital twins feed reliability models with simulated operating data [11], are validated against physical experiments [12], and are discussed as maintenance tools whose suitability is still open [42]. Multi-fidelity fusion combines scarce physical tests with abundant simulation for mission-reliability prediction, weighting the simulation by its fidelity gap [13], and simulators generate test scenarios for autonomous systems [25, 14]. Model verification, validation and uncertainty quantification identify extrapolation to untested conditions as a major challenge [26]. *What is missing:* a rule that states which assumption about the simulator is needed before its output may reduce a zero-failure test, and a measured sim–real discordance that makes the assumption checkable.

(c) *Simulation-augmented evaluation of perception.* The simulation-to-reality gap of camera object detectors has been measured across day, night, fog and rain [16] and for drone detection [17]; synthetic images are used to test driving perception, with a distribution mismatch that biases test results [27], and perception digital twins reproduce field data only approximately [43]. Statistical model checking evaluates perception in simulation only [44], simulation performance can fail to predict field performance [18], and a statistical foundation for aligning synthetic and real testing outcomes has been called for [15]. Closest to our question, Sim2Val uses simulator outputs as control variates to reduce the number of real samples needed for a confidence bound on mean performance (CoRL 2025, arXiv preprint) [45], SCAPE combines limited real rollouts with simulation proxies for policy evaluation (preprint) [46], and multi-fidelity fusion weights simulation by its fidelity gap in reliability prediction [13]; none treats the zero-failure regime. Prediction-powered inference and its stratified variant combine many proxy outcomes with few real labels through a paired rectifier [28, 29]. *What is missing:* these methods estimate a mean performance or reduce its variance; none addresses the zero-failure regime, where the quantity to bound is a rare failure probability and the rectifier itself would cost almost n_0 real events per unit.

(d) *Finite-sample guarantees and assurance for learned perception.* Conformal prediction gives distribution-free finite-sample guarantees [47], calibrated risk control extends them to general losses [30], and they have been applied to object detection [31], keypoint-based pose estimation [48], perception for safe navigation [32] and, in this journal, anomaly detection in cyber-physical systems [33]. Reliability of machine-learning components is also modelled directly, through multi-version architectures [7], safety-assurance

121 cases for object detectors [8] and failure-free-operation definitions under an
 122 operational profile [9]. *What is missing:* these guarantees are calibrated on
 123 real data from the deployment distribution; they do not say how much of
 124 that calibration data may be synthetic, nor how a guarantee transfers to a
 125 unit whose own data are absent.

126 3. Demonstration setting

127 *Unit, demand and failure.* The unit under test is a fixed camera s with a
 128 fixed detector. The camera observes *events*: maximal runs of frames that
 129 contain the target object. It samples frames with period K and an unknown
 130 phase φ , and runs the detector on the sampled frames. An event is *missed*
 131 ($M = 1$) if no sampled frame is a hit; with H the hit frames of an event and
 132 φ uniform, $\mathbb{P}(M = 1) = 1 - |H \bmod K|/K$. The miss probability of camera s
 133 is p_s , and the demonstration goal is the assurance claim $p_s \leq \varepsilon$ at confidence
 134 $1 - \delta$. $U(x, n)$ denotes the one-sided $(1 - \delta)$ Clopper–Pearson upper bound
 135 for x failures in n demands. We call an accepted assurance claim of this form
 136 a *certificate*. In the language of acceptance sampling, δ is the consumer’s
 137 risk; the producer’s risk is reported as the power of a plan at the observed
 138 miss rate. All claims concern the miss probability averaged over the uniform
 139 sampling phase; a camera deployed with one fixed, unknown phase is covered
 140 only on average over phases.

141 *Simulated evidence and twins.* A simulator produces data $Y \sim Q$. A *paired*
 142 *twin* renders a synthetic counterpart of each real event and runs the same
 143 detector at the same sampling phase, giving M' . We write $q = \mathbb{P}(M' = 1)$
 144 and $D = \mathbb{P}(M = 1, M' = 0)$ (real miss, twin catch), the *discordance*. Every
 145 rate in this paper is per event; with three sprite variants per event, the twin
 146 catches an event if at least two variants do, and an event counts as discordant
 147 if it is discordant at one or more sampling phases (worst phase; an integer
 148 count, conservative with respect to the phase-averaged rate, which we give
 149 in parentheses). Table 2 collects the notation.

150 *Random model.* Scenes are drawn from a population; given a scene, events
 151 are drawn i.i.d. from the scene’s event law and each receives an independent
 152 uniform phase; given the event and the phase, the real and twin outcomes
 153 are determined (fixed detector, fixed sprite seeds). Every per-event indicator
 154 below—real miss, twin miss, discordance at a random phase, and the worst-
 155 phase indicator—is therefore a Bernoulli variable, independent across events

Table 2: Notation.

symbol	meaning
ε, δ	target miss probability; allowed probability of a false acceptance
$n_0(\varepsilon, \delta)$	zero-failure sample size $\lceil \ln \delta / \ln(1 - \varepsilon) \rceil$
$K, \varphi, d_{\min}$	sampling period, sampling phase, minimum event length (frames)
M, M'	real miss, twin miss of an event at the same phase
p_s, q_s, D_s	real miss probability, twin miss probability, discordance $\mathbb{P}(M = 1, M' = 0)$ of camera s
m, n_s, N	calibration scenes, paired events of scene s , $N = \sum_s n_s$
X, J	discordant events; scenes with at least one discordant event
$U(x, n)$	one-sided $(1 - \delta)$ Clopper–Pearson upper bound
fleet bound, population bound	$U(X, N)$ and $U(J, m)$ (Result 4)
OP*, OP-A, OP-C	operating points $(\varepsilon, d_{\min}, K) = (20\%, 32, 16)$, $(20\%, 32, 8)$, $(10\%, 32, 1)$

156 given the scene. In the audit, the “truth” of a scene is its empirical event
 157 law, resampled with replacement.

158 4. What simulated evidence can buy: four results

159 The four results below are stated in words; Appendix A gives the formal
 160 statements, the assumptions and the proofs, each with a numerical check.

161 *Result 1 (no free credit).* Suppose the simulated data are independent of the
 162 real demands and no assumption links the simulator to reality, so that for
 163 every simulator law Q both a perfect unit ($p = 0$) and a unit just worse
 164 than the target ($p \downarrow \varepsilon$) are possible. Then any acceptance procedure that
 165 is valid at confidence $1 - \delta$ accepts a perfect unit with probability at most
 166 $\delta / (1 - \varepsilon)^n$ when it uses n real demands, whatever the simulator and however
 167 much simulation it uses. Accepting a perfect unit with certainty therefore
 168 still needs $n \geq n_0(\varepsilon, \delta)$ real demands. The reason is that the simulated
 169 data have the same law under both hypotheses, while an all-success real
 170 sample is possible under both. This makes precise, for the success-run test,
 171 the principle that external evidence cannot shorten a test whose validity

Table 3: Result 2: exact credit c versus the continuous form c^* ($\varepsilon = \delta = 5\%$, $n_0 = 59$), and the two-tier plan run with the twin event pool as an independent simulator (worst case $p = \varepsilon$; closed form given the pass probability, and 100,000 Monte-Carlo runs).

δ'	7.5%	10%	15%	20%
exact credit c (demands)	8	14	22	27
saving w.r.t. n_0	14%	24%	37%	46%
continuous form c^*	7	13	21	27
false acceptance, twin pool (closed form)	7.3%	9.9%	14.8%	19.1%
Monte Carlo	7.2%	10.1%	14.8%	19.3%

172 must hold for every external law [19], and the related limit on combining
 173 evidence for ultra-high dependability [20]. A paired twin is built from the
 174 real events and is therefore *not* an independent simulator; pairing is what
 175 Result 3 exploits.

176 *Result 2 (the exact price of relaxing confidence).* If the demonstration may
 177 accept a worse unit with probability $\delta' > \delta$ when the simulator is wrong—
 178 that is, if one accepts a higher consumer’s risk—the real demands can be
 179 reduced by at most

$$c = n_0(\varepsilon, \delta) - n_0(\varepsilon, \delta'),$$

180 and this reduction is attained by a two-tier plan: run the zero-failure test with
 181 $n_0(\varepsilon, \delta')$ demands when a simulator test passes and with $n_0(\varepsilon, \delta)$ otherwise.
 182 The plan keeps confidence $1 - \delta$ only if a simulator of a worse-than-target
 183 unit passes with probability at most π^* : 6.1%, 3.0%, 1.5% and 1.0% for
 184 $\delta' = 7.5, 10, 15$ and 20% at $\varepsilon = \delta = 5\%$. This requirement is itself a
 185 link between simulator and reality, so Result 2 is best read as the price of
 186 lowering the confidence, not as a free credit. The frequently used continuous
 187 form $\lfloor \ln(\delta'/\delta)/(-\ln(1 - \varepsilon)) \rfloor$ is only a lower bound on c . Table 3 gives the
 188 numbers.

189 Monte-Carlo runs match the closed form within 0.29 percentage points on
 190 a 40-point grid. Used as an independent simulator (its events resampled in-
 191 dependently of the real draws), the twin event pool passes the simulator test
 192 in 98.2% of runs, the credit is almost fully realised, and the false-acceptance
 193 probability at $p = \varepsilon$ is δ' up to the pass probability: the credit is bought ex-
 194 actly with the extra error. An off-the-shelf synthetic benchmark (BMC [49];
 195 30 events long enough for the operating point) never passes at $\varepsilon = 5\%$ (0%):

196 it has too few synthetic events to bound its own miss probability below 5%.
 197 At $\varepsilon = 20\%$ it passes in every run and buys the same credit as the twin
 198 (supplementary Table S13), which shows that passing the simulator test is
 199 no evidence of fidelity. Monte-Carlo standard errors are about 0.1%.

200 *Result 3 (paired-twin bound).* For any pairing of real and twin events, the real
 201 miss probability is at most the twin miss probability plus the discordance,

$$p \leq q + D,$$

202 with no assumption on the simulator (the bound splits the real misses into
 203 those the twin also misses and those it catches). On a single camera the
 204 bound does not save demands: if $q \leq q_U$ holds with confidence $1 - \delta/2$ and
 205 $D \leq \varepsilon - q_U$ is demonstrated by a zero-discordance run at level $\delta/2$, the
 206 camera needs $\lceil \ln(\delta/2)/\ln(1 - \varepsilon + q_U) \rceil$ paired events, i.e. 72, 91 and 122
 207 for $q_U = 0, 1, 2\%$ at $\varepsilon = \delta = 5\%$, all above $n_0 = 59$. A twin built with a
 208 pessimistic (shorter) event length remains valid: its miss probability can only
 209 grow and its discordance only shrink. On the fixed-camera twin, shortening
 210 every event by 50% raises the twin miss rate at OP* from 0.87% to 1.67%.
 211 With a uniform phase the discordance probability of an event is $|R' \setminus R|/K$,
 212 where R, R' are the residues modulo K of the real and twin hit frames; the
 213 worst-phase indicator $\mathbf{1}\{R' \setminus R \neq \emptyset\}$ dominates it, so the bounds of Result 4
 214 also hold for the integer worst-phase counts we report.

215 *Result 4 (transfer across scenes).* Let m calibration scenes contribute $N =$
 216 $\sum_s n_s$ paired events, X of them discordant, and let J scenes show at least one
 217 discordant event. (a) *Population bound (the transfer guarantee).* If the scenes
 218 are drawn independently from a scene population (events within a scene may
 219 be dependent), the population-average discordance of a new camera is at
 220 most $U(J, m)$ with confidence $1 - \delta$, so its expected miss probability is at
 221 most its expected twin miss probability plus $U(J, m)$. The bound counts
 222 scenes, not events; it needs no independence of events within a scene. Its
 223 price is the number of scenes: with $J = 0$, $U \leq 10\%$ needs 29 scenes and
 224 $U \leq 5\%$ needs 59. With the 78 fixed-camera scenes of this study, $J = 0, 1, 2, 3$
 225 give $U(J, m) = 3.8\%, 5.9\%, 7.9\%$ and 9.6% at level δ and $4.6\%, 6.9\%, 9.0\%$
 226 and 10.8% at level $\delta/2$: a single discordant scene costs about two percentage
 227 points, and a bound below $\varepsilon/2 = 10\%$ tolerates three discordant scenes at
 228 level δ but only two at $\delta/2$. Adding events to existing scenes does not tighten
 229 it; only new scenes do. (b) *Fleet bound (calibrated cameras, independent*

Table 4: Data needed for the per-camera form of Result 4(c) with no discordance ($J = 0$): minimum number of scenes $m_{\min}$, events per scene n at $m = m_{\min}$ and at $m = 2m_{\min}$, total events, and the asymptotic floor $\ln(1/\delta)/(\gamma t)$.

t	γ	$m_{\min}$	n at $m_{\min}$	n at $2m_{\min}$	total at $2m_{\min}$	floor
2.5%	5%	59	183	28, 3,304	2,397	
2.5%	10%	29	158	28, 1,624	1,198	
5%	5%	59	91	14, 1,652	1,198	
5%	10%	29	78	14, 812	599	
10%	5%	59	44	7, 826	599	
10%	10%	29	38	7, 406	300	

events). For the calibrated cameras, the event-weighted discordance is at most $U(X, N)$ with confidence $1 - \delta$ (for $\delta \leq 1/4$), however heterogeneous the cameras are; hence their event-weighted miss rate is at most their twin miss rate plus $U(X, N)$. Applying it requires a separate bound on the twin miss rate, with its own share of δ ; where real misses exist, direct Clopper–Pearson on them is at least as informative. The proof compares the Poisson–binomial count with a binomial of the same mean [50] and uses a lower bound on the probability that a binomial reaches its mean [51], applied to the complementary count. This bound assumes that events are independent given the scenes, so positive dependence within a scene makes it optimistic, and it says nothing about a new camera. (c) *Per-camera limit*. A per-camera statement, “ $p \leq q + t$ except on a set of cameras of mass γ ”, follows if every scene has at least $n_{\min}$ events, with $\gamma = U(J, m)/(1 - (1 - t)^{n_{\min}})$. Table 4 turns this into a data requirement: for $\gamma = 5\%$ at least 59 scenes, each with tens to hundreds of events. No public benchmark we know offers this, so the per-camera form is reported as a limit.

Demonstration procedure. The procedure evaluated in Section 6 is: (1) declare the domain and pair every real event of m calibration scenes from it with its twin; count discordant scenes J ; (2) *domain rule*: if $U_{\delta}(J, m) > \varepsilon/2$, withdraw the transfer claim for the whole domain (a design rule for go/no-go, not a theorem); (3) for a new camera of the declared domain, run its twin until its own miss probability is bounded, q_U , at level $\delta/2$ (a twin *run* is one rendering of one synthetic event in one sprite variant; runs are treated as independent, which is optimistic when runs share a trajectory); (4) accept the camera with no real demand if $q_U + U_{\delta/2}(J, m) \leq \varepsilon$, otherwise fall back to the zero-failure test (Fig. 1). By Result 4(a) and a union bound, with confidence $1 - \delta$ the population-average discordance is at most $U_{\delta/2}(J, m)$.

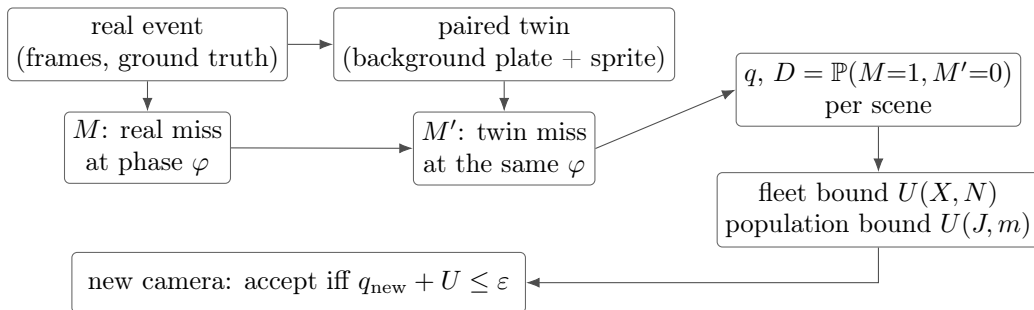


Figure 1: The demonstration with a paired twin. Each real event is paired with a synthetic twin that replaces the object and keeps the scene; both are evaluated by the same detector at the same sampling phase. Discordances calibrated across scenes bound the miss probability of a new camera through $p \leq q + D$.

257 Among *accepted* cameras the average discordance can be larger, by at most
 258 the factor $1/\mathbb{P}(\text{accept})$, because acceptance selects cameras with a low twin
 259 miss rate; the average miss probability of accepted cameras is therefore at
 260 most $\varepsilon - U_{\delta/2} + U_{\delta/2}/\mathbb{P}(\text{accept})$. At OP^* , where 97.4% of the static cam-
 261 eras are accepted, the inflation is 0.2 percentage points; out of domain it
 262 is unbounded (Section 6.3). The acceptance is *not* a per-camera guarantee.
 263 In the audit we nevertheless count, per camera, acceptances with $p_s > \varepsilon$
 264 (“false acceptances”) as a diagnostic of how the average guarantee behaves
 265 on individual cameras.

266 5. Experimental protocol

267 *Data.* The corpus is rebuilt from the raw detections and pixel ground truth
 268 of CDnet 2014 [52] and LASIESTA [53] (frames in true temporal order),
 269 with the synthetic BMC benchmark [49] as an off-the-shelf simulator; the
 270 detector-defined corpus has 3,170 events from 101 real videos. For the twin
 271 we use ground-truth events: a frame is a *hit* if a detector box covers at least
 272 half of the largest ground-truth component or overlaps its bounding box with
 273 $\text{IoU} \geq 0.3$, and an event counts if its largest object has at least 256 pixels
 274 (primary definition; the original definition keeps every ground-truth event).
 275 Scenes are split into tiers before any twin result: a *static* (main) domain of
 276 78 scenes (30 CDnet, 48 LASIESTA; 232 events of any duration, 180 of at
 277 least 32 frames), a night tier (6 scenes), a turbulence tier (4 scenes; only 2
 278 were run and they hold 2 events, so the tier is reported but not certified), and

279 4 pan-tilt-zoom scenes that are excluded because the twin assumes a static
 280 background.

281 *Detector and twin.* The detector is a fixed single-stage segmentation network
 282 (YOLO family [54]; yolo26s-seg [55], confidence 0.25, input 640 px, CPU).
 283 Per scene, a background plate is the per-pixel median of up to 60 frames
 284 without ground-truth foreground. For each real event and each of three sprite
 285 variants (fixed seeds), the twin removes the real object (mask dilated by 3
 286 px), fills it from the plate, and draws a synthetic object on the ground-truth
 287 trajectory: a vehicle cut from Virtual KITTI 2 [56], chosen by aspect ratio,
 288 or a procedurally dressed human rendered with MPFB [57] with a four-frame
 289 walking cycle mirrored to the motion direction. The sprite height equals the
 290 ground-truth box height, its width is clamped to $\pm 10\%$ of the box width, its
 291 brightness follows the frame luminance, and a soft shadow is added. Only the
 292 object is synthetic; scene, illumination and background clutter are real. The
 293 detector runs on twin frames until the twin hits cover all residues modulo 48,
 294 which is exact for every K dividing 48. The median detector confidence on
 295 twin hits is 0.83 (real: 0.90). Building the twins of the 78 static scenes took
 296 18.3 CPU-hours (median 0.13 per scene) on a laptop CPU.

297 *What the twin consumes.* The paired twin uses, per camera, empty frames
 298 (for the plate) and the annotated trajectories of the camera’s real events (for
 299 sprite placement). Calibration therefore costs annotated events: 180 at OP*,
 300 i.e. 2.3 per static camera when amortized over the domain. For a *new* camera,
 301 the exchange rates below assume that the twin miss rate q_{new} can be measured
 302 without that camera’s real events, e.g. with synthetic trajectories drawn on
 303 its empty background. We did not build such trajectory-free twins; we use
 304 the paired q_s in their place. The discordance of a trajectory-free twin has not
 305 been calibrated, so the reported savings are conditional on the paired-twin
 306 discordance carrying over, and are upper bounds.

307 *Operating points.* An operating point is $(\varepsilon, d_{\min}, K)$ with events of at least
 308 $d_{\min}$ frames. The primary point OP* = (20%, 32, 16) at $\delta = 5\%$ was fixed by
 309 a rule chosen before the twin was built (the largest K for which direct certifi-
 310 cation needs at most 150 events at 80% power and 44 scenes give a population
 311 bound of at most $\varepsilon/2$), using the pooled real miss rate of all CDnet events
 312 (11.3% over 161 events, 112 events for direct certification). In the static
 313 domain the real miss rate at OP* is only 0.73%; we report this, keep OP*,
 314 and add the sensitivity points OP-A (20%, 32, 8) and OP-C (10%, 32, 1) and

the full grid $d_{\min} \in \{1, 4, 8, 12, 16, 24, 32\}$, $K \in \{1, 2, 3, 4, 6, 8, 12, 16, 24, 48\}$. The night and turbulence tuning criteria and their held-out scenes were fixed before tuning. These choices are dated in internal project records but were not publicly registered. Four choices were made after results were available and are disclosed: the primary event definition (largest object at least 256 pixels) was adopted after the initial twin run, with the original definition kept as a co-reported sensitivity (Table 12); the LASIESTA scene list was rebuilt (20 to 48 scenes) after correcting the frame order of that dataset, before the static-domain twin was built; the two held-out turbulence scenes were not run; and the post-hoc point (16, 16) of Section 6 was selected after inspecting the static-domain results. The operating point OP^* , the tiers, the audit design and the night criterion were fixed before the static-domain twin results; everything else in Section 6 is exploratory.

Why a paired twin. A reweighting alternative [58] needs overlap between synthetic and real events. A classifier separating BMC events from real events reaches a median AUC of 0.64 with temporal features, 0.94 with geometry and 0.77 with detector confidence alone (all targets); the gap lies mostly in geometry, so a reweighted synthetic benchmark cannot stand in for the real event distribution, and the twin is matched on geometry by construction.

Audit. Leave-one-scene-out (LOSO): each scene is certified with the bound computed from the other scenes. Monte Carlo (10,000 draws): calibration scenes are resampled from the static domain and a new camera is drawn from it; a false certificate is a certified camera with $p_s > \varepsilon$ (the resampled calibration set may contain the test camera, which makes this check optimistic; the LOSO audit does not have this defect); the “truth” of a camera is its own event pool, exact over phases. Two must-fail scenarios are run (Section 6.3). All randomness uses seed 42.

6. Results

6.1. Certificates by tier

In the static domain, 3 of 180 events at OP^* are discordant (1.67%; phase-averaged 0.3125 events, 0.17%), all in one scene; hence $U(X, N) = 4.25\%$ and, with $J = 1$ of 78 scenes, $U(J, m) = 5.94\%$ (Table 5, Fig. 2). Both are below $\varepsilon/2 = 10\%$. At night, 12 of 25 events are discordant (48%; phase-averaged 8.21) and 5 of 6 scenes; pooling all tiers raises the bounds to 11.5% and 14.7%.

Table 5: Certificates by tier at OP* ($\delta = 5\%$). Discordant events are counted at the worst sampling phase (in parentheses: phase-averaged count, mean over the three sprite variants; per-scene tables use the majority rule, which gives 8.31 at night); $\hat{p}$, $\hat{q}$: real and twin miss rates averaged over the phases / at the worst phase (any fixed deployment phase); J = scenes with a discordant event.

tier	events	$\hat{p}$ avg/worst	$\hat{q}$ avg/worst	discordant	$U(X, N)$	J/m	$U(J, m)$
static (main)	180	0.7%/2.2%	0.9%/2.2%	3 (0.3125)	4.25%	1/78	5.94%
night	25	36.2%/52.0%	4.8%/16.0%	12 (8.21)	65.9%	5/6	99.1%
turbulence	2	6.2%/50.0%	0.0%/0.0%	1	97.5%	1/2	97.5%
all tiers	207	5.1%/8.7%	1.3%/3.9%	16	11.5%	7/86	14.7%

350 6.2. Exchange rate

351 Fig. 3 and Table 7 show the real events a static camera saves when the
352 twin replaces direct certification, across 70 operating points. A camera is
353 certified with no real event when its twin miss rate plus the transfer bound
354 (both at $\delta/2$, the bound computed without the camera) stays below ε . At
355 OP* the direct test needs only $n_0 = 14$ events, because the static-domain
356 miss rate is low; a twin run to convergence saves those 14 events at 76 of 78
357 cameras, at a measured cost of 0.044 CPU-hours per saved event (conditional
358 on the discordance carrying over, Section 5; no accepted camera has $p_s > \varepsilon$
359 at OP*, but acceptances were not audited at every grid cell). **The twin as**
360 **built saves no real events; the saving requires about 27 twin runs**
361 **per camera.** With three runs per event, as built, the twin’s own miss rate
362 cannot be bounded tightly enough, and at most 7 cameras save anything
363 at any grid point. The saving grows with the real miss rate, because direct
364 certification becomes expensive: it reaches hundreds of events per camera
365 when $\hat{p}$ exceeds 10%. At the post-hoc point (16, 16) it is 22 events per camera
366 (Table 6). The target $\varepsilon = 20\%$ is a demonstrator, not a safety target. At
367 $\varepsilon = 5\%$ the transfer term alone is $U_{\delta/2}(0, 78) = 4.6\%$ even with no discordant
368 scene, leaving 0.4% for the twin’s own miss rate, i.e. at least 967 miss-free
369 twin runs per camera; with 78 scenes the twin route is therefore impractical
370 at $\varepsilon = 5\%$, and 59 or more scenes with no discordance are needed before
371 it can work. For the calibrated cameras themselves, direct Clopper–Pearson
372 on their pooled real misses gives 5.0%, close to the fleet discordance bound
373 4.25%: the fleet certificate adds little where real data exist, and its role is as
374 the building block of the transfer.

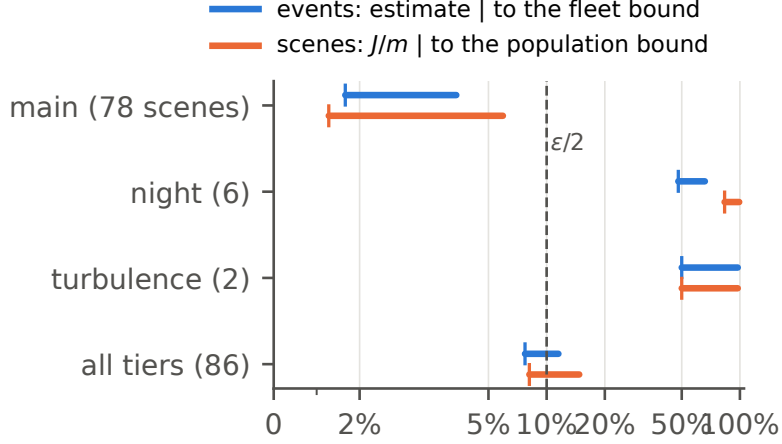


Figure 2: Certificates by tier at OP* (event unit, worst phase). Blue: discordant events per event and the fleet bound; orange: discordant scenes per scene and the population bound. Dashed: $\varepsilon/2$.

6.3. Audit and must-fail scenarios

Is the certificate right because it is empty? Table 8 cross-tabulates real and twin outcomes. In the static domain at OP* almost every event is caught by both: 4 events have a real miss at some phase, in 2 scenes, and the twin also misses 76% of the expected real misses, i.e. one event. The zero-discordance result is therefore mostly a statement about a detector that rarely misses in this domain, not a test of the twin. At the post-hoc point the twin sees most real misses; at night it sees almost none.

LOSO and Monte Carlo. At OP*, 1 of 78 scenes has a discordance above the bound computed without it (Fig. 5, Table 9); the Monte-Carlo false-certification rate is 0.0%. Because real misses are rare at OP*, this audit cannot reveal much: no static scene has $p_s > \varepsilon$ there, so a false certificate is impossible by construction and the 0% rate carries no evidence. The violating scenes are expected under a population-average guarantee. The informative test is must-fail 2 below, which can fail and does.

Must-fail 1: temporal. At $K = 48$, 4 static scenes have $p_s > \varepsilon$ and the procedure refuses them in 100% of runs; but most of these misses are temporal (events shorter than K , which the twin reproduces by construction), and only 2 scenes exceed ε through detector misses (Table 10, top).

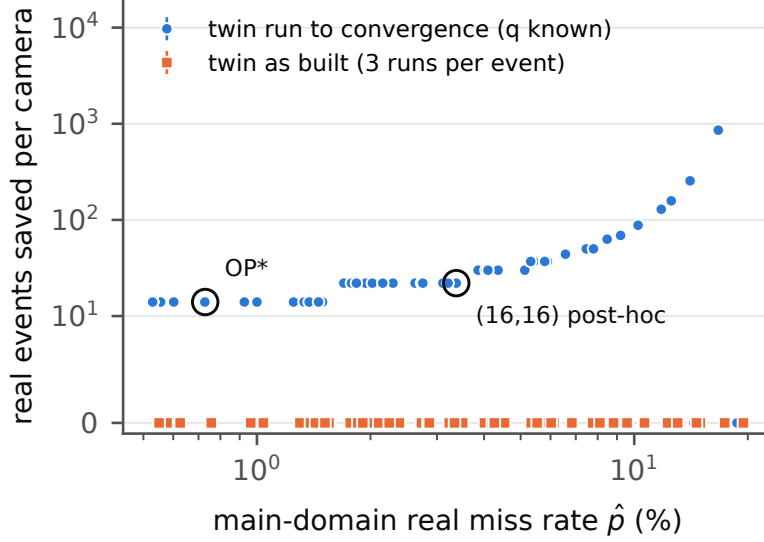


Figure 3: Exchange rate: real events saved per static camera (median and interquartile range over 78 cameras) against the static-domain real miss rate, one point per $(d_{\min}, K)$ at $\varepsilon = 20\%$. Blue: twin run to convergence; orange: twin as built (three runs per event). Vertical axis: symmetric log (linear below 10). Circled: OP^* and the post-hoc point $(16, 16)$.

394 *Must-fail 2: detector.* We add k of the 6 night scenes to the calibration
395 as if they were static-domain scenes; at OP^* and $(16, 16)$ every real miss
396 of these scenes is a detector miss. The transfer bound rises with k (Fig. 4,
397 Table 10, bottom): from 5.9% to 13.6% at OP^* , crossing $\varepsilon/2$ for some subsets
398 at $k = 3$ and for all at $k = 4$, so the domain-level certificate is withdrawn; no
399 static camera is ever falsely certified (0). *Out-of-domain pass-through:* the
400 per-camera rule does not protect the night cameras themselves. The night
401 twin misses almost nothing (its appearance model only dims the sprite), so
402 each night camera, judged with a bound computed from the static scenes,
403 is certified; 4 of 6 night cameras have $p_s > \varepsilon$ and are falsely certified at
404 OP^* (3 at $(16, 16)$; Table 10, middle). Each night miss rate rests on 2 to
405 10 events and is uncertain; the finding is that the acceptance rule admitted
406 these cameras although their observed miss rates exceed ε and their twins
407 miss almost nothing, not a precise estimate of their miss rates.

Table 6: Exchange rate at four operating points (78 static cameras). n_{dir} : real events of the smallest direct plan (failures allowed, Clopper–Pearson acceptance) with 80% power at the pooled static-domain $\hat{p}$; the zero-failure plan alone needs n_0 events; saved: median real events saved per camera by a twin run to convergence; runs: twin runs needed per camera; CPU-h: CPU-hours per saved real event.

point	$\hat{p}$	n_{dir}	saved	cams	runs	CPU-h
OP* (20%, 32, 16)	0.73%	14	14	76	27	0.044
OP-A (20%, 32, 8)	0.56%	14	14	77	23	0.036
OP-C (10%, 32, 1)	0.56%	29	29	77	68	0.052
(16, 16) post hoc	3.38%	22	22	75	39	0.041
worst-phase convention (valid for any fixed phase):						
OP*	2.2%	22	22	74	27	–
(16, 16) post hoc	10.0%	82	82	73	39	–
twin as built: cameras saving = 4, 5, 0, 2 (median saving 0)						

408 6.4. Illustrative example at a post-hoc operating point

409 To see the certificate under actual real misses, we moved, after inspecting
410 the data, to the nearest operating point with at least 20 missed real events:
411 $(d_{\min}, K) = (16, 16)$. This choice is post hoc and is reported as an illustration
412 only. There, 20 of 200 events have a real miss at some phase and the twin
413 also misses 13 of them (counting events); weighting by phases, the twin
414 reproduces 71% of the expected real misses; $J = 3$, $U(J, m) = 9.6\%$, 2 of 78
415 scenes exceed the LOSO bound, and no camera is falsely certified (0).

416 6.5. Fixed-phase and site-level checks

417 *Worst phase.* A deployed camera keeps one sampling phase. Counting
418 an event as missed if it is missed at any phase, for the real event, the twin
419 (majority of variants) and the discordance alike, keeps the bound valid for
420 every fixed phase, because an event missed at some phase is either missed by
421 the twin or discordant at that phase. At OP* the static-domain real and twin
422 miss rates become 2.2% and 2.2% (phase-averaged 0.7% and 0.9%), while the
423 discordance counts and both bounds are unchanged (Table 5). The direct
424 plan then needs 22 events, and a twin run to convergence accepts 74 of 78
425 cameras, saving 22 events each (Table 6); 1 accepted camera has a worst-
426 phase real miss rate above ε (the camera whose three discordant events define
427 J), so under the fixed-phase reading the audit is no longer clean.

Table 7: Operating-point grid at $\varepsilon = 20\%$ (condensed; full grid in the supplement). Each cell: static-domain real miss rate $\hat{p}$ / real events for direct certification / median real events saved per camera by a twin run to convergence. The twin as built saves none at every cell. OP* is (32, 16).

$d_{\min} \backslash K$	1	8	16	24	48
32	0.6%/14/14	0.6%/14/14	0.7%/14/14	1.4%/14/14	5.8%/37/0
24	0.5%/14/14	0.7%/14/14	1.5%/14/14	2.7%/22/22	8.5%/63/0
16	1.0%/14/14	1.8%/22/22	3.4%/22/22	5.4%/37/37	12.2%/141/0
12	1.5%/14/14	2.3%/22/22	3.9%/30/30	5.9%/37/37	12.6%/158/0
8	1.5%/14/14	2.7%/22/22	5.1%/30/30	7.5%/50/50	14.4%/288/0
4	1.8%/22/22	5.8%/37/37	9.2%/69/69	11.8%/129/129	18.7%/5925/0
1	2.2%/22/22	10.2%/88/88	14.1%/255/255	16.7%/858/858	23.4%/-/0

Table 8: Real versus twin outcomes (expected event counts under a uniform sampling phase; twin = majority of three variants). Recall: share of expected real misses that the twin also misses. ^{ph} post hoc.

	static, OP*	static, (16, 16) ^{ph}	night, OP*
real miss, twin miss	1.00	4.81	0.75
real miss, twin catch (D)	0.31	1.94	8.31
real catch, twin miss	0.56	3.62	0.44
real catch, twin catch	178.12	189.62	15.50
events	180	200	25
events missed at some phase	4	20	13
twin recall of real misses	76%	71%	8%

428 *Sites.* Scenes filmed at the same site share background, viewpoint and
429 illumination, so the scene may be too fine a unit of independence. Grouping
430 the static scenes into 22 sites (CDnet challenge categories and LASIESTA
431 sequence groups, a coarse proxy for sites), 1 site shows a discordant event at
432 OP*, and the population bound becomes $U(J_{\text{site}}, m_{\text{site}}) = 19.8\%$, above $\varepsilon/2$:
433 under the domain rule the transfer claim would be withdrawn at site level.
434 At (16, 16) it is 25.9% (2 of 22 sites). A leave-one-site-out audit finds 1 site
435 above its bound at OP* and 2 at (16, 16). The scene-level results therefore
436 hold only if scenes, not sites, are the independent units.

437 6.6. Failure modes

438 At night the twin sees few real misses: it predicts 8% of the expected
439 missed events, and the tier bound is 99.1%. Fig. 6 shows why: the appearance
440 model scales sprite brightness with the scene, but has no headlights, glare or
441 sensor noise. An attempt to fix this was specified in advance: night-specific
442 appearance profiles were tuned on four night scenes and judged on two held-

Table 9: Leave-one-scene-out: scenes whose discordance D_s exceeds the bound U_{-s} computed without them. None is falsely certified, because $p_s \leq \varepsilon$.

point	scene	n	D_s	p_s	q_s	U_{-s}	$p_s > q_s + U_{-s}$
OP*	traffic	8	3.9%	3.9%	0.0%	3.8%	yes
(16, 16) ^{ph}	traffic	13	9.6%	9.6%	0.0%	8.0%	yes
(16, 16) ^{ph}	tramCrossroad_1fps	4	9.4%	12.5%	6.2%	8.0%	no

Table 10: Must-fail scenarios. Top: temporal ($d_{\min} = 1$, $K = 48$), static scenes with $p_s > \varepsilon$, real misses split into detector and temporal misses; all refused in every Monte-Carlo run. Middle: night scenes at OP* (every miss is a detector miss), and whether each is falsely certified when judged alone. Bottom: population bound when k night scenes leak into the calibration set (median over subsets; withdrawn = share of subsets with a bound above $\varepsilon/2$).

<i>1: temporal</i>						
scene	n	p_s	detector	temporal	q_s	refused
port_0_17fps	5	99.6%	43.3%	56.2%	100.0%	100%
traffic	13	53.4%	24.5%	28.8%	37.2%	100%
tramCrossroad_1fps	11	77.7%	6.1%	71.6%	74.2%	100%
tunnelExit_0_35fps	34	83.6%	5.2%	78.4%	87.9%	100%
<i>2: detector (night)</i>						
scene	n	p_s	q_s	D_s	disc.	false cert.
bridgeEntry	2	15.6%	0.0%	15.6%	1	no
busyBoulevard	4	96.9%	0.0%	96.9%	4	yes
fluidHighway	4	26.6%	0.0%	26.6%	2	yes
streetCornerAtNight	10	28.1%	11.9%	20.6%	4	yes
tramStation	2	0.0%	0.0%	0.0%	0	no
winterStreet	3	33.3%	0.0%	33.3%	1	yes
k leaked	0	1	2	3	4	6
U_{pop}	5.9%	7.8%	9.4%	10.1%	10.8%	13.6%
withdrawn	0%	0%	0%	50%	100%	100%

out scenes, with success defined as a median held-out per-scene discordance of at most 5%. The best profile barely moved the tuning discordance (47.2% to 47.0%, phase-averaged) and the held-out median was 23.2% on 14 events, so the criterion failed, tuning was stopped, and night is declared outside the domain. The turbulence tier holds 2 events at OP* and is not certified.

6.7. Sensitivity

Event definition. Keeping every ground-truth event (295 instead of 232 static events of any duration) leaves OP* almost unchanged: 182 events, 3 discordant, $U(X, N) = 4.20\%$, $J = 1$ (Table 12). At the post-hoc point the

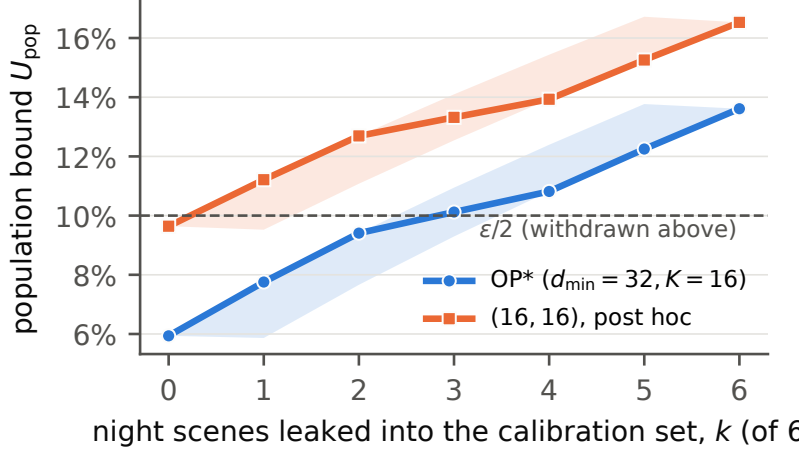


Figure 4: Must-fail 2: the population bound when k night scenes leak into the static calibration set (median over the $\binom{6}{k}$ subsets, band: min-max). The domain certificate is withdrawn above $\varepsilon/2$. Individual night cameras are still certified when judged alone (out-of-domain pass-through, text).

original definition adds small, hard objects, and the population bound rises to 11.4% with $J = 4$, above $\varepsilon/2$: the illustrative example depends on the definition, while OP* does not. *Operating point.* At OP-A and OP-C the static-domain miss rate is 0.56% and 0.56%, and no static-domain event is discordant. *Twin details.* The majority rule over sprite variants (1, 2 or 3 of 3) does not change any static-domain count. Table 11 compares twin and real hit rates on the same frames: the twin is slightly pessimistic at every aspect error except one vehicle bin, and both hit rates fall when the ground-truth box has an unusual aspect (occlusion, objects entering the frame). Clamping the sprite width, which affects 70% of twin frames (more than the frames with an aspect error above 10% in Table 11, because 1-pixel rounding also triggers the clamp), lowers the twin hit rate by only 3.0 points more than the real hit rate on the same frames (631 event-variant pairs): it inflates q , not D . The median detector confidence on hits is lower for the twin than for real objects, especially for persons. Full tables are in the supplement.

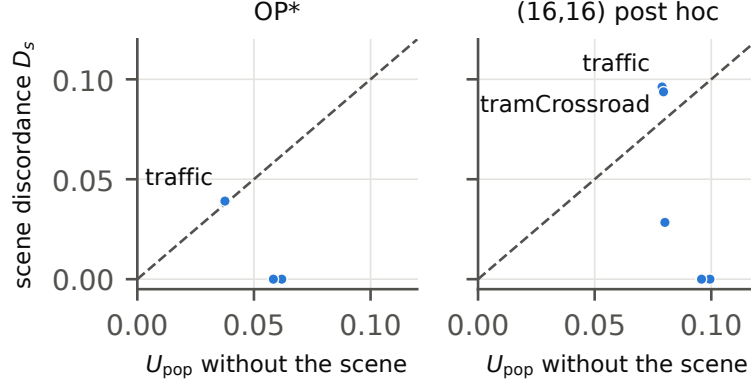


Figure 5: Leave-one-scene-out: per-scene discordance against the population bound computed without the scene. Points above the diagonal exceed the bound; the proposition bounds the population mean, not each scene.

7. Discussion and limitations

A narrow window. Simulation pays in events only when direct certification is expensive, that is, when the real miss rate is not far below ε ; there the twin must also be accurate, and discordances make the transfer bound grow. When the miss rate is low, n_0 is small and there is little to save. The grid (Table 7) shows both ends: 14 events at OP^* , hundreds at operating points where $\hat{p}$ approaches ε .

Time. The waiting time saved at camera s is $n_{\text{saved}}/\lambda_s$, where λ_s is the event rate in deployment; we do not estimate it, because the benchmarks publish no frame rates and their clips are dense in events. If one CPU-hour is valued like one hour of waiting, the twin breaks even when $\lambda_s < 1/c \approx 23$ events per hour at OP^* , with c the CPU-hours per saved event; this equivalence is an assumption, not a measurement.

Domain. The transfer bound is a population-average guarantee: it withdraws the domain certificate when out-of-domain scenes enter the calibration, but it does not detect an out-of-domain camera. The domain must therefore be declared and checked by other means (for example, illumination statistics), and a per-camera guarantee requires the data of Result 4(c). Three items are left for future work: a twin that needs no annotated real trajectory of the new camera, with its own calibrated discordance; a pre-specified in-domain stress test with cameras whose real miss rate exceeds ε ; and a



Figure 6: Synthetic person sprites (CC0) on a neutral background under the brightness gain used for day and for night scenes. Dimming alone does not reproduce night imaging, which explains the low twin miss rate at night.

488 domain check (for example a per-camera discordance sentinel on a few real
489 events) to stop out-of-domain pass-through.

490 *Data and twin.* No static scene has 59 or more qualifying events, so per-
491 camera ground truth is the scene’s own finite event pool; the twin builds its
492 plate from real empty frames and therefore needs a static background; the
493 sprite library lacks elevated viewpoints for vehicles; and we did not use a full
494 3-D simulator. The operating point was fixed from the pooled miss rate of all
495 CDnet events, which turned out to be much higher than the static-domain
496 rate; we kept it rather than re-select after seeing the twin results. Scenes are
497 not independent draws: several LASIESTA sequences share a room or set-
498 up, and the site-level bound of Section 6.5 exceeds $\varepsilon/2$. The cost comparison
499 counts real events and CPU-hours only; annotating the calibration events and
500 building plates are additional costs. Finally, an event miss is a component-
501 level failure; linking it to a system hazard requires a hazard analysis of the
502 application.

503 *Independence and phase.* The results assume independent events given
504 the scene and independent scenes. Events of one camera share a background,
505 a detector and, in deployment, a single sampling phase; benchmark scenes are
506 not a random sample of deployments. Correlation of events within a scene
507 does not invalidate the population bound of Result 4(a) or the leave-one-
508 scene-out audit, because both count scenes, not events: a scene contributes
509 one indicator whatever the dependence among its events. It does affect the

Table 11: Twin fidelity on the same frames (static domain): hit rates by object kind and sprite aspect error before the width clamp, and median detector confidence on hits. Negative differences make the twin pessimistic.

kind	aspect error	frames	twin hit	real hit	diff. (pts)
person	$\leq 10\%$	8,870	91.7%	94.8%	-3.2
person	10-20%	5,899	90.1%	94.2%	-4.2
person	20-40%	4,404	74.0%	86.2%	-12.2
person	$> 40\%$	5,449	34.4%	44.5%	-10.0
vehicle	$\leq 10\%$	4,105	86.9%	89.0%	-2.1
vehicle	10-20%	2,569	88.0%	90.2%	-2.2
vehicle	20-40%	3,087	87.6%	85.1%	+2.5
vehicle	$> 40\%$	3,881	64.9%	74.2%	-9.3
person	confidence		0.86	0.92	
vehicle	confidence		0.74	0.77	

Table 12: Event-definition sensitivity (static domain). Primary: largest object ≥ 256 px; original: every ground-truth event. Discordant events at the worst phase; LOSO: scenes above the bound computed without them.

definition	point	N	$\hat{p}$	disc.	$U(X, N)$	$J / U(J, m)$	LOSO
primary	OP*	180	0.73%	3	4.25%	1 / 5.9%	1
	OP-A	180	0.56%	0	1.65%	0 / 3.8%	0
	OP-C	180	0.56%	0	1.65%	0 / 3.8%	0
	(16, 16) ^{ph}	200	3.38%	12	9.54%	3 / 9.6%	2
original	OP*	182	0.72%	3	4.20%	1 / 5.9%	1
	OP-A	182	0.55%	0	1.63%	0 / 3.8%	0
	OP-C	182	0.55%	0	1.63%	0 / 3.8%	0
	(16, 16) ^{ph}	205	4.73%	13	9.89%	4 / 11.4%	0

^{ph} post-hoc point.

510 fleet bound of Result 4(b), which counts events and becomes optimistic under
511 positive within-scene dependence; the population bound needs only indepen-
512 dence across scenes. The main results certify the phase-averaged miss rate;
513 the worst-phase convention of Section 6.5 covers any fixed phase at a higher
514 real-event cost.

515 *Threats to validity.* The audit tables and figures use level δ , while ac-
516 ceptance uses $\delta/2$ for the transfer term; at $\delta/2$ the bounds are larger (Re-
517 sult 4(a)). Software: Python 3.11.9, NumPy 1.26.4, SciPy 1.16.3, pandas
518 2.3.3, Ultralytics 8.4.7, OpenCV 4.10.0; CPU: Intel Core i7-1185G7. The
519 detector, the event definition and the hit rule are fixed; a different detector
520 changes p , q and D but not the theorems. The number of scenes, not the
521 number of events, limits the transfer certificate: the static domain reaches
522 $U(J, m) = 5.94\%$ with one discordant scene among 78, and more scenes are

needed for tighter bounds. The audit at OP^* is weak because real misses are rare there; the post-hoc point shows the mechanism under real misses but was selected after inspecting the data, and its population bound depends on the event definition (Table 12).

Practical guidance. The results suggest a simple decision rule for practitioners who consider replacing a field demonstration with a twin. (i) Compute n_0 at the target (ε, δ) and estimate the real miss rate; if it is far below ε , the direct test is already short and a twin can save at most n_0 events. (ii) If a twin is built, declare its domain before calibration and calibrate discordance on as many scenes of that domain as possible; the number of scenes sets the resolution (Result 4(a)). (iii) Run each new camera’s twin long enough to bound its own miss rate; three runs per event, as here, are not enough. (iv) Do not use the transfer certificate to admit a camera whose conditions differ from the calibration domain: the certificate is withdrawn when such scenes enter the calibration, but a single such camera passes through.

8. Conclusion

Without an assumption linking simulation to reality, synthetic evidence buys no real events in zero-failure certification, and relaxing the error level buys an exactly computable number. A paired twin turns the question into one of discordance. The population bound is the transfer guarantee: it counts scenes, tolerates dependence within a scene, and its price is the number of scenes; the fleet bound covers only the calibrated cameras and assumes independent events. At site level the population bound of this study exceeds $\varepsilon/2$. Measured on public benchmarks, the exchange rate is modest and domain-bound, and the transfer certificate lets out-of-domain cameras pass through unless the domain is declared.

CRediT authorship contribution statement

Lam Quoc Dat: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The author declares no competing interests. This research received no specific grant from any funding agency in the public, commercial, or not-for-

556 profit sectors.

557 Declaration of generative AI and AI-assisted technologies in the 558 manuscript preparation process

559 During the preparation of this work the author used an AI assistant
560 (Claude, Anthropic) to write analysis and plotting scripts, to check the num-
561 bers of the manuscript against the result files, and to draft and edit text.
562 After using this tool, the author reviewed and edited the content as needed
563 and takes full responsibility for the content of the publication.

564 Data availability

565 Code, event manifests, result tables and the CC0 person-sprite set are
566 publicly available at [https://github.com/lamquocdat01/no-free-credi](https://github.com/lamquocdat01/no-free-credit-ress-artifact)
567 [t-ress-artifact](https://github.com/lamquocdat01/no-free-credit-ress-artifact) (tag `ress-v1.0`, commit `b60195a56383`). Raw video frames
568 (CDnet 2014, LASIESTA, BMC) and Virtual KITTI 2 sprites are not redis-
569 tributed under their licences; the repository provides download instructions
570 and SHA-256 checksums to regenerate every derived file.

571 Appendix A. Formal statements and proofs

572 This appendix states the results of Section 4 formally and proves them.
573 Theorem A.1 is Result 1, Theorem A.3 is Result 2, Lemma A.5 and Propo-
574 sition A.15 are Result 3, and Proposition A.11, Theorem A.9 and Proposi-
575 tion A.13 are Result 4(a), (b) and (c).

576 Appendix A.1. Setting and notation

577 **Events and misses.** A camera (scene) s produces events $i = 1, 2, \dots$ (a
578 maximal run of frames containing the target object). The system samples
579 frames periodically with period K and an unknown phase $\varphi \in \{0, \dots, K-1\}$;
580 a detector is run on the sampled frames. An event is *missed* ($M = 1$)
581 when no sampled frame of the event is a hit (the detector finds the object).
582 With $H \subseteq \mathbb{Z}$ the hit frames of the event and φ uniform, $\mathbb{P}(M = 1 \mid H) =$
583 $1 - |H \bmod K|/K$, where $H \bmod K$ is the set of residues of H . The miss
584 rate of camera s is $p_s = \mathbb{P}(M = 1)$ over its events and phases. Target: certify
585 $p_s \leq \varepsilon$ with confidence $1 - \delta$.

586 **Clopper–Pearson bound.** For $x \in \{0, \dots, n\}$, $F_{n,p}(x) = \mathbb{P}(\text{Bin}(n, p) \leq x)$
587 and $U(x, n) = U_\delta(x, n)$ is the one-sided $(1 - \delta)$ upper Clopper–Pearson bound
588 [59]: the solution u of $F_{n,u}(x) = \delta$ for $x < n$, and 1 for $x = n$. Since $F_{n,p}(x)$
589 is strictly decreasing in p , $U(x, n) < p \iff F_{n,p}(x) < \delta$, and U is non-
590 decreasing in x . Validity: $\mathbb{P}(U(\text{Bin}(n, p), n) < p) \leq \delta$ for every p .

591 **Success run.** $n_0(\varepsilon, \delta) = \lceil \ln \delta / \ln(1 - \varepsilon) \rceil$ is the smallest n with $(1 - \varepsilon)^n \leq \delta$
592 (e.g. $n_0(5\%, 5\%) = 59$, $n_0(20\%, 5\%) = 14$).

593 **Simulator and twin.** Simulator data Y have a law Q on an arbitrary space.
594 A *twin* is a paired simulator: every real event i has a synthetic counterpart
595 with miss indicator M'_i under the same sampling phase. $q = \mathbb{P}(M' = 1)$, and
596 the *discordance* is $D = \mathbb{P}(M = 1, M' = 0)$ (real miss, twin catch).

597 Appendix A.2. C1: no free credit

598 A certification procedure is a (possibly randomised) test $\phi(X, Y) \in \{0, 1\}$
599 with real data $X = (M_1, \dots, M_n) \sim \text{Bern}(p)^{\otimes n}$ independent of simula-
600 tor data $Y \sim Q$. It is *valid at level δ on a class $\mathcal{P}$* of pairs (p, Q) if
601 $\sup_{(p, Q) \in \mathcal{P}, p > \varepsilon} \mathbb{P}_{p, Q}(\phi = 1) \leq \delta$.

602 **Theorem A.1 (C1).** Assume $\mathcal{P}$ places no restriction linking Q and p : for
603 every Q there are $(0, Q) \in \mathcal{P}$ and $(p, Q) \in \mathcal{P}$ for all $p \in (\varepsilon, \varepsilon + \eta)$ for some
604 $\eta > 0$. If ϕ is valid at level δ on $\mathcal{P}$, then for every Q and every simulator
605 sample size,

$$(1 - \varepsilon)^n \mathbb{P}_{0, Q}(\phi = 1) \leq \delta.$$

606 In particular, certifying a perfect system ($p = 0$) with probability one requires
607 $n \geq n_0(\varepsilon, \delta)$ real events, whatever the amount of simulator data; power $1 - \beta$
608 at $p = 0$ requires $n \geq \ln(\delta / (1 - \beta)) / \ln(1 - \varepsilon)$.

609 *Proof.* Fix Q and write $g(y) = \mathbb{E}[\phi(0^n, y)]$ (expectation over the internal
610 randomisation). Under $p = 0$, $X = 0^n$ almost surely, so $\mathbb{P}_{0, Q}(\phi = 1) =$
611 $\mathbb{E}_Q[g(Y)]$. Under $p > \varepsilon$, the event $\{X = 0^n\}$ has probability $(1 - p)^n$ and is
612 independent of Y , hence $\mathbb{P}_{p, Q}(\phi = 1) \geq (1 - p)^n \mathbb{E}_Q[g(Y)] = (1 - p)^n \mathbb{P}_{0, Q}(\phi =$
613 $1)$. Validity gives $(1 - p)^n \mathbb{P}_{0, Q}(\phi = 1) \leq \delta$ for all $p \in (\varepsilon, \varepsilon + \eta)$; letting $p \downarrow \varepsilon$
614 proves the inequality. With $\mathbb{P}_{0, Q}(\phi = 1) = 1$ it reads $(1 - \varepsilon)^n \leq \delta$, i.e. $n \geq n_0$;
615 with $\mathbb{P}_{0, Q}(\phi = 1) \geq 1 - \beta$ it gives the power bound. $\square$

616 *Remark A.2.* The argument only uses that the simulator data have the same
617 law under the two hypotheses and that the all-success real sample has positive
618 probability under both. It is the success-run analogue of the observation

619 that external data cannot increase power under strict type-I error control
620 over all external-data laws [19]. *Check.* The zero-failure test with $n = 59$ at
621 $p = \varepsilon = 5\%$: $(0.95)^{59} = 0.0485$ (MC 0.0491, 10^5 runs).

622 *Appendix A.3. C2: the price of borrowing*

623 Suppose one accepts a two-tier guarantee: level δ when the simulator is
624 “right” and only level $\delta' > \delta$ in general.

625 **Theorem A.3** (C2, exact credit). (i) *Converse.* Under the assumption of
626 *Theorem A.1*, any procedure valid at level δ' on $\mathcal{P}$ that certifies a perfect
627 system with probability one uses $n \geq n_0(\varepsilon, \delta')$ real events. Hence the credit,
628 the number of real events saved with respect to $n_0(\varepsilon, \delta)$, is at most

$$c = n_0(\varepsilon, \delta) - n_0(\varepsilon, \delta').$$

629 (ii) *Attainment.* Let $T(Y) \in \{0, 1\}$ be any simulator test (“the simulator
630 passes”) and let B be the procedure: if $T = 1$ run the zero-failure test with
631 $n_1 = n_0(\varepsilon, \delta')$ real events, otherwise with $n_0 = n_0(\varepsilon, \delta)$. Then B is valid at
632 level δ' on $\mathcal{P}$, saves exactly c events whenever $T = 1$, and is valid at level δ
633 on the subclass

$$\mathcal{P}_0 = \left\{ (p, Q) : p > \varepsilon \Rightarrow \mathbb{P}_Q(T = 1) \leq \pi^* \right\}, \quad \pi^* = \frac{\delta - (1 - \varepsilon)^{n_0}}{(1 - \varepsilon)^{n_1} - (1 - \varepsilon)^{n_0}},$$

634 i.e. where a simulator of a system that is truly worse than ε rarely passes.

635 (iii) *Continuous lower bound.* $c \geq c^* = \lfloor \ln(\delta'/\delta)/(-\ln(1 - \varepsilon)) \rfloor$.

636 *Proof.* (i) Theorem A.1 at level δ' gives $(1 - \varepsilon)^n \leq \delta'$, whose smallest integer
637 solution is $n_0(\varepsilon, \delta')$. (ii) For $p > \varepsilon$, with $\pi = \mathbb{P}_Q(T = 1)$ and T independent
638 of the real data, $\mathbb{P}(B = 1) = \pi(1 - p)^{n_1} + (1 - \pi)(1 - p)^{n_0} \leq (1 - \varepsilon)^{n_1} \leq \delta'$,
639 and $\leq \delta$ as soon as $\pi \leq \pi^*$ (the expression is affine in π and decreasing in
640 p). (iii) Write $a = \ln \delta / \ln(1 - \varepsilon)$, $a' = \ln \delta' / \ln(1 - \varepsilon)$ and $r = a - a' =$
641 $\ln(\delta'/\delta)/(-\ln(1 - \varepsilon))$. Since $n_0(\delta) \geq a$ and $n_0(\delta') < a' + 1$, $c > r - 1$; c being
642 an integer, $c \geq \lfloor r \rfloor = c^*$. $\square$

643 *Remark A.4* (numbers and checks). At $\varepsilon = \delta = 5\%$: $\delta' = 7.5, 10, 15, 20\%$
644 give $c = 8, 14, 22, 27$ ($-14\%, -24\%, -37\%, -46\%$ of 59). The exact form is
645 maximal on all 40 grid points ($(1 - \varepsilon)^{n_0(\delta')} \leq \delta' < (1 - \varepsilon)^{n_0(\delta')-1}$: yes); MC
646 (10^5 runs per point) matches the closed form within 0.29 percentage points.
647 *Empirical C2* (procedure B , T = the one-sided CP bound of the simulator’s

miss rate $\leq \varepsilon$, 10^5 runs, $\varepsilon = 5\%$, $d_{\min} = 32$, $K = 1$): with the paired twin of 78 static scenes (180 events) the simulator passes in 98.2% of runs; realised credit 7.9/13.7/21.6/26.5 events and false certification at the worst case $p = \varepsilon$ equal to 7.2%/10.1%/14.8%/19.3% for $\delta' = 7.5/10/15/20\%$ — the credit is bought exactly with the extra error $\delta' - \delta$. The off-the-shelf synthetic benchmark (30 events of length ≥ 32) never passes (0.0%): too few synthetic events to bound its own miss rate below 5%.

Appendix A.4. C3: the twin lemma

Lemma A.5 (C3). *For any pairing (M, M') of a real event and its twin, $p \leq q + D$ with $D = \mathbb{P}(M = 1, M' = 0)$. No assumption links the twin to reality.*

Proof. $p = \mathbb{P}(M = 1, M' = 1) + \mathbb{P}(M = 1, M' = 0) \leq \mathbb{P}(M' = 1) + D = q + D$. $\square$

Corollary A.6 (a per-camera twin does not save events). *If $q \leq q_U$ holds with confidence $1 - \delta/2$ and $D \leq \varepsilon - q_U$ is certified for the same camera by a zero-discordance success run at level $\delta/2$, the number of paired real events is $\lceil \ln(\delta/2) / \ln(1 - (\varepsilon - q_U)) \rceil = 72, 91, 122$ for $q_U = 0, 1, 2\%$ at $\varepsilon = \delta = 5\%$, all larger than $n_0 = 59$. The twin can only help when D is calibrated once, on many scenes, and transferred (C4).*

Remark A.7 (event unit). With a uniform phase, $x_i = \mathbb{P}_\varphi(M_i = 1, M'_i = 0) = |R'_i \setminus R_i|/K$, where R_i, R'_i are the residue sets of the real and twin hit frames of event i (twin catch decided by the majority of three sprite variants). The *worst-phase* indicator $W_i = \mathbf{1}\{R'_i \setminus R_i \neq \emptyset\}$ satisfies $\mathbb{P}(M_i = 1, M'_i = 0 \mid \text{event}) = x_i \leq W_i$. All certificates below are stated for Bernoulli indicators Z_i ; they hold verbatim with $Z_i = W_i$ and then bound the (larger) worst-phase discordance rate. This integer form is the one used for the reported certificates.

Appendix A.5. fleet bound: event-weighted certificate for the calibrated fleet

Lemma A.8 (mean–median). *For $N \geq 2$ and $0 < p < 1 - 1/N$, $F_{N,p}(\lfloor Np \rfloor) > 1/4$. Consequently, if $\delta \leq 1/4$ and $F_{N,p}(x) < \delta$, then $x \leq Np - 1$.*

Proof. Greenberg and Mohri [51] prove $\mathbb{P}(B \geq \mathbb{E}B) > 1/4$ for $B \sim \text{Bin}(N, r)$ with $r > 1/N$. Apply it to $B = N - \text{Bin}(N, p) \sim \text{Bin}(N, 1 - p)$ with $1 - p >$

681 $1/N$: $\mathbb{P}(\text{Bin}(N, p) \leq Np) > 1/4$, and $\{\text{Bin}(N, p) \leq Np\} = \{\text{Bin}(N, p) \leq$
682 $\lfloor Np \rfloor\}$. For the consequence: $F_{N,p}$ is non-decreasing, so $F_{N,p}(x) < \delta \leq$
683 $1/4 < F_{N,p}(\lfloor Np \rfloor)$ forces $x < \lfloor Np \rfloor$, i.e. $x \leq \lfloor Np \rfloor - 1 \leq Np - 1$. Check:
684 $\inf F_{N,p}(\lfloor Np \rfloor)$ over $N = 2..2000$ and a fine grid of p is 0.2500, approached
685 at $N = 2$, $p \rightarrow 1/2$ (the constant $1/4$ is tight). $\square$

686 **Theorem A.9** (fleet bound). *Let $Z_1, \dots, Z_N$ be independent, $Z_i \sim \text{Bern}(d_i)$*
687 *with arbitrary $d_i \in [0, 1]$ (heterogeneous across scenes and events), $X =$*
688 *$\sum_i Z_i$ and $\bar{d} = \frac{1}{N} \sum_i d_i$. If $\delta \leq 1/4$ and $\bar{d} < 1 - 1/N$, then*

$$\mathbb{P}(U(X, N) < \bar{d}) \leq \delta.$$

689 *With $d_i = D_{s(i)}$ (the discordance of the scene of event i), $\bar{d} = \bar{D}_w =$*
690 *$\sum_s n_s D_s / N$, the event-weighted discordance of the calibrated cameras; com-*
691 *bined with Lemma A.5, $\bar{p}_w \leq \bar{q}_w + U(X, N)$ with probability at least $1 - \delta$.*

692 *Proof.* If $\bar{d} = 0$ there is nothing to prove. Otherwise let $x^* = \max\{x :$
693 $F_{N,\bar{d}}(x) < \delta\}$ (if the set is empty the failure probability is 0). By monotonicity
694 of U , $\{U(X, N) < \bar{d}\} = \{X \leq x^*\}$. By Lemma A.8, $x^* \leq N\bar{d} - 1$. Hoeffding's
695 Theorem 4 [50]: for a sum S of independent Bernoulli variables with mean
696 $N\bar{d}$ and $0 \leq b \leq N\bar{d} - 1$, $\mathbb{P}(S \leq b) \leq F_{N,\bar{d}}(b)$. With $b = x^*$, $\mathbb{P}(X \leq x^*) \leq$
697 $F_{N,\bar{d}}(x^*) < \delta$. $\square$

698 *Remark A.10* (what it does not say; checks). The guarantee is conditional on
699 the calibrated scenes and weighted by events: it does not bound any single
700 camera and does not transfer to a new camera (if scenes are random, X is
701 over-dispersed; e.g. $D_s \in \{0, 1\}$ gives $\mathbb{P}(X = 0) = (1 - \mu)^m \gg (1 - \mu)^N$).
702 *Checks.* Six synthetic heterogeneity patterns on a fixed configuration of 44
703 scenes with $N = 304$ events (a calibration set from an earlier stage of the
704 study, used only as a test bed for the theorem): exact Poisson–binomial
705 failure ≤ 0.0471 , MC (10^5) ≤ 0.0473 . Measured discordance probabilities
706 of the study (event level, main domain, five operating points, as measured
707 and rescaled to mean 2% and 5%; 15 cases): exact failure ≤ 0.0126 , MC
708 ≤ 0.0129 , all $\leq \delta = 0.05$.

709 *Appendix A.6. population bound: transfer to a new camera*

710 **Proposition A.11** (population bound). *Let scenes $S_1, \dots, S_m$ be i.i.d. from*
711 *a scene population, each with $n_s \geq 1$ paired events whose discordance indica-*
712 *tors are independent $\text{Bern}(D_{S_s})$ given the scene, $\mu = \mathbb{E}[D_S]$, and $J = \#\{s :$*

713 $X_s \geq 1\}$ with X_s the number of discordant events in scene s . Then

$$\mathbb{P}(U(J, m) < \mu) \leq \delta.$$

714 For a new camera drawn from the same population, $\mathbb{E}[p_{\text{new}}] \leq \mathbb{E}[q_{\text{new}}] +$
 715 $U(J, m)$ with probability $\geq 1 - \delta$ over the calibration, and $\mathbb{P}(D_{\text{new}} > t) \leq$
 716 $U(J, m)/t$ (Markov).

717 *Proof.* Let $B_s = \mathbf{1}\{X_s \geq 1\}$. Given the scene, $\mathbb{P}(B_s = 1 \mid S_s) = 1 -$
 718 $(1 - D_{S_s})^{n_s} \geq D_{S_s}$ because $(1 - D)^n \leq 1 - D$ for $n \geq 1$. Hence $\beta_s :=$
 719 $\mathbb{P}(B_s = 1) \geq \mu$, and the B_s are independent across scenes. By a coupling
 720 $(B_s = \mathbf{1}\{V_s \leq \beta_s\} \geq \mathbf{1}\{V_s \leq \mu\})$ with i.i.d. uniforms V_s , J stochastically
 721 dominates $\text{Bin}(m, \mu)$. Since $U(\cdot, m)$ is non-decreasing, $\mathbb{P}(U(J, m) < \mu) \leq$
 722 $\mathbb{P}(U(\text{Bin}(m, \mu), m) < \mu) \leq \delta$ by Clopper–Pearson validity. The corollary
 723 follows from Lemma A.5 applied to the new camera and averaged over the
 724 population. $\square$

725 *Remark A.12* (price and checks). The resolution is set by the number of
 726 scenes: with $J = 0$, $U \leq 10\%$ needs $m \geq 29$ and $U \leq 5\%$ needs $m \geq 59$.
 727 *Checks.* The tight case $D_S \in \{0, 1\}$, $n_s = 1$, $m = 78$ (then $J \sim \text{Bin}(m, \mu)$
 728 exactly), $\mu \in \{1, 2, 5, 10, 20\}\%$, 10^5 runs: maximum failure 0.0401. Leave-
 729 one-scene-out on the 78 static scenes at the post-hoc challenge point ($d_{\min} =$
 730 16 , $K = 16$): $D_s > U(J_{-s}, m - 1)$ for 2 of 78 scenes (the proposition bounds
 731 the population mean, not each scene).

732 *Appendix A.7. per-camera form: a per-camera (quantile) certificate and its*
 733 *data requirement*

734 **Proposition A.13** (per-camera form). *Under the assumptions of Proposi-*
 735 *tion A.11 with $n_s \geq n_{\min}$ for all s , let $t \in (0, 1)$, $\pi_t = \mathbb{P}(D_S > t)$ and*
 736 *$h_t = 1 - (1 - t)^{n_{\min}}$. Then*

$$\mathbb{P}(\pi_t > U(J, m)/h_t) \leq \delta.$$

737 Hence, with $\hat{\gamma} = U(J, m)/h_t$, a new camera satisfies $p_{\text{new}} \leq q_{\text{new}} + t$ except
 738 on a set of cameras of population mass at most $\hat{\gamma}$, with confidence $1 - \delta$.

739 *Proof.* $\mathbb{P}(B_s = 1) \geq \mathbb{P}(D_S > t)(1 - (1 - t)^{n_s}) \geq \pi_t h_t$. The coupling argument
 740 of Proposition A.11 gives $J \succeq_{\text{st}} \text{Bin}(m, \pi_t h_t)$, so $\mathbb{P}(U(J, m) < \pi_t h_t) \leq \delta$,
 741 which is the claim. $\square$

742 *Remark A.14* (data requirement; checks). With $J = 0$, $\hat{\gamma} \leq \gamma$ requires $m \geq$
743 $m_{\min}$ with $1 - \delta^{1/m_{\min}} \leq \gamma$ ($m_{\min} = 59$ for $\gamma = 5\%$, 29 for $\gamma = 10\%$) and $n_{\min} \geq$
744 $\ln(1 - U(0, m)/\gamma)/\ln(1 - t)$ events in every scene; the total $m n_{\min}$ decreases
745 to the floor $\approx \ln(1/\delta)/(\gamma t)$ as m grows. *Check*: two-point populations $D_S \in$
746 $\{0, t'\}$, $t' > t$, $m = 78$, 10^5 runs: maximum failure 0.0286.

747 *Appendix A.8. C5a: a pessimistic-length twin is conservative under periodic*
748 *sampling*

749 **Proposition A.15** (C5a). *Let an event occupy frames $F = \{o, \dots, o + d - 1\}$*
750 *and let the detector outcome on each frame be fixed (a deterministic function*
751 *of the frame). For the truncated event $F_L = \{o, \dots, o + L - 1\}$, $L \leq d$,*
752 *with hit set $H_L = H \cap F_L$, the miss probability under a uniform phase,*
753 *$m(L) = 1 - |H_L \bmod K|/K$, is non-increasing in L . Consequently a twin built*
754 *with a pessimistic (shorter) length L has $q_L \geq q$ and discordance $D_L \leq D$,*
755 *so the C3 bound $q_L + D_L$ remains valid and the miss term is conservative.*

756 *Proof.* $L \leq L'$ implies $H_L \subseteq H_{L'}$, hence $H_L \bmod K \subseteq H_{L'} \bmod K$ and
757 $m(L) \geq m(L')$. A twin catch at phase φ for the truncated twin implies
758 a catch for the full twin, so $\{M = 1, M'_L = 0\} \subseteq \{M = 1, M' = 0\}$ and
759 $D_L \leq D$; Lemma A.5 holds for any pairing. $\square$

760 *Remark A.16* (check). On the twin of the 78 static scenes, shortening every
761 event by $x \in \{10, 25, 50\}\%$ gives non-decreasing q for every event and every
762 $K \in \{1, 4, 8, 16, 48\}$ (all monotone: yes); e.g. at $d_{\min} = 32$, $K = 16$, q rises
763 from 0.87% to 1.67% at $x = 50\%$.

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